

● COMPUTER NETWORKS · MASTER STUDY SERIES



MASTER EDITION

Master Revision & Exam-Ready Book

The whole course in one volume: course map, formula vault, protocol tables, command arsenal, subnetting drills, a full mock exam with answer key, and a glossary of every term.

Course Map

Formula Vault

Protocol Master Tables

Command Arsenal

Subnetting Drills

Mock Exam + Key

Glossary

HOW TO USE THIS BOOK

1 week out: read the Course Map + every lecture's Rapid Revision Card. **3 days out:** drill the Formula Vault and Subnetting Drills daily. **1 day out:** sit the Mock Exam under time pressure, then check the key. **Morning of:** skim the Glossary and Command Arsenal. That's the whole system.

STUDY-READY EDITION

COVERS LECTURES 1–13 · TERM 5 · SST

1 The Course Map — 13 Lectures at a Glance

Lec	Topic	Core idea to never forget	Lec	Topic	Core idea to never forget
1	Intro & Architecture	Edge vs core; packet switching wins by sharing; throughput = min link	8	Routing II	DV shares tables (RIP), LS shares maps (OSPF), BGP between ASes
2	Packets & Layers	Encapsulation: headers added down, stripped up; MACs change, IPs don't	9	DNS & HTTP	Cache chain → root→TLD→auth; HTTP stateless, cookies add memory
3	Subnetting I	Network = IP AND mask; usable = $2^h - 2$; private ranges	10	TCP / UDP	SYN SYN-ACK ACK; reliability = ACKs + retransmit + window
4	Subnetting II	hosts+2 → 2^N → prefix; VLSM largest-first; NAT = port table	11	ARP·NAT·DHCP	DORA; broadcast Q / unicast A; NAT table = port discriminator
5	Graphs I	BFS=queue=hops (RIP); Dijkstra=heap=cost (OSPF)	12	Troubleshooting	Climb the OSI ladder; the toolset tells you where it breaks
6	Graphs II	Bellman-Ford survives negative edges; MST ↔ STP	13	The Journey	9 phases from click to pixels — every lecture appears once
7	Routing I	Longest Prefix Match; TTL -1 per hop; hop-by-hop			

2 The Formula Vault

Delay & performance (L1, L5)

```

d_nodal = d_proc + d_queue + d_trans + d_prop
d_trans = L / R          L bits, R bps
d_prop = d / s           s ≈ 2×108 m/s
ρ = (a · L) / R          traffic
intensity < 1!
throughput = min(R1...Rn)
store-fwd over N links = N × (L/R)

```

Subnetting (L3, L4)

```

total addresses = 2(32 - N)
usable hosts    = 2(32 - N) - 2
network        = IP AND mask
broadcast      = host bits → all 1s
subnets created = 2(new - old prefix)
need H hosts   → smallest 2h ≥ H+2
                → prefix = 32 - h
last-octet jumps: /25→128 /26→192 /27→224
                  /28→240 /29→248 /30→252

```

Routing (L7, L8)

```

LPM: most bits matched wins
OSPF cost = 100 Mbps / link BW
          100M→1 · 10M→10 · 1M→100
DV update: min(mine, to-nbr + nbr's)
RIP: hops ≤ 15, 16 = ∞
BGP: LOCAL_PREF → AS_PATH len → MED

```

Transport & TTL (L2, L10)

```

ports: 0-1023 well-known
       22 SSH · 53 DNS · 80 HTTP
       443 HTTPS · 25 SMTP · 3306 MySQL
TCP: seq = byte count
     ack = next byte expected
     win = receiver buffer
TTL: 64 Linux/Mac · 128 Windows
TIME_WAIT = 2×MSL ≈ 60 s
UDP header 8 B · TCP ≥ 20 B
hops ≈ sending TTL - received TTL

```

3 Protocol Master Tables

Model	Layers						
OSI (7)	Application	Presentation	Session	Transport	Network	Data Link	Physical
TCP/IP (4)	Application			Transport	Internet	Network Access	
Data unit	message/data			Segment	Packet	Frame	bits
Addressing	—			ports	IP	MAC	—

Protocol	Layer	Job	Memorable fact
ARP	L2	IP → MAC locally	broadcast question, unicast answer
DHCP	App (UDP 67/68)	IP + mask + GW + DNS	DORA — request stays broadcast
DNS	App (UDP 53)	names → IPs	13 root servers A–M; cache by TTL
HTTP(S)	L7	web transfer	stateless; cookies add memory; TLS = L6
TCP	L4	reliable ordered bytes	3-way open, 4-way close, fast retransmit on 3 dup-ACKs
UDP	L4	fast datagrams	8-byte header; DNS/video/games/QUIC
ICMP	L3	control & errors	ping, Time Exceeded, Fragmentation Needed
IP	L3	end-to-end delivery	best-effort; TTL loop-guard; fragmentation
Ethernet	L2	local frames	64–1518 B; FCS silent-discard
RIP	IGP	DV routing	max 15 hops; 30 s updates; split horizon
OSPF	IGP	LS routing	flood LSAs → LSDB → Dijkstra; areas off Area 0
BGP	EGP	between ASes	path-vector; AS_PATH loop check; policy rules
TLS	L6	encryption	TLS 1.3 = 1-RTT; DH forward secrecy; SNI
STP	L2	switch loop-stopper	builds spanning tree; blocks redundant links

4 The Command Arsenal

DIAGNOSE

```
ip addr · ip link
ping -c 4 host
tracert host
dig +short host
dig +trace host
dig -x IP
ss -tulpn
ip route show
ip neigh show
```

TERMINAL

```
interfaces + MACs
L3 reach + RTT
path + latency map
quick DNS
full walk
reverse
listeners + procs
routing table
ARP cache
```

FIX & CAPTURE

```
ip route add NET via NH
ip neigh flush all
sudo tcpdump -n port 53
sudo tcpdump -w f.pcap
sudo tcpdump -A port 80
curl -v http://host
mtr host
```

TERMINAL

```
static route
clear ARP
watch DNS
record
read HTTP
HTTP probe
live path stats
```

5 Subnetting Drill (do all 10, then check)

The problems

1. 10.20.30.40/16 → network? broadcast?
2. 192.168.1.77/26 → network? usable range?
3. 172.16.90.77/20 → mask? network? usable count?
4. Need 50 hosts → prefix? usable?
5. Need 500 subnets from /16 → new prefix? hosts each?
6. How many /28s in a /24?
7. 172.25.10.5 — public or private?
8. 172.32.10.5 — public or private?
9. 192.168.10.130/25 → network? broadcast?
10. 10.45.60.100/20 → network? broadcast?

Answers

1. 10.20.0.0 · 10.20.255.255
2. 192.168.1.64 · .65–.126 (bc .127)
3. 255.255.240.0 · 172.16.80.0 · 4094
4. /26 · 62
5. /25 · 126 each ($2^9=512$ subnets)
6. 16
7. Private (172.16–31)
8. Public — outside /12!
9. 192.168.10.128 · .255
10. 10.45.48.0 · 10.45.63.255

The 30-second method — worked on drill #3 (172.16.90.77/20)

1. **Block size from the octet that changes:** /20 → third octet keeps 4 prefix bits → block = $2^{8-4} = 16$ → boundaries 0, 16, 32, 48, 64, **80**, 96...
2. **Network:** largest boundary ≤ 90 → third octet 80 → **172.16.80.0**.
3. **Broadcast:** next boundary – 1 → 96 – 1 → **172.16.95.255**.
4. **Usable:** .80.1 – .95.254 → host bits = 12 → $2^{12} - 2 = 4094$.

TRAPS THAT COST MARKS

"Usable" excludes network + broadcast (**–2**, always). 172.**32** is public (outside 16–31). /31 and /32 have no usable hosts. Count **subnets** with 2^{borrowed} , **hosts** with $2^{\text{remaining}} - 2$ — don't mix them.

SANITY CHECKS

Network + block ≤ next network. Broadcast = network + block – 1. First usable = network + 1. If your broadcast crosses the next boundary, your block size is wrong — recount the borrowed bits.

6 Mock Exam — 90 minutes, 70 marks

Section A — MCQ (1 × 10 = 10 marks)

1. Throughput on a path with links 100/10/50 Mbps equals: (a) 100 (b) 50 (c) 10 (d) 53.3 MCQ
2. Which is NOT a transport-layer duty? (a) ports (b) reliability (c) MAC lookup (d) flow control MCQ
3. EtherType 0x0806 identifies: (a) IPv4 (b) ARP (c) IPv6 (d) VLAN MCQ
4. Usable hosts in /23: (a) 512 (b) 510 (c) 1022 (d) 254 MCQ
5. Dijkstra cannot handle: (a) cycles (b) negative weights (c) sparse graphs (d) weighted edges MCQ
6. OSPF's algorithm on the LSDB: (a) Bellman-Ford (b) BFS (c) Dijkstra (d) Kruskal MCQ
7. The URL part never sent to servers: (a) path (b) query (c) fragment (d) host MCQ
8. Three duplicate ACKs trigger: (a) slow start (b) window 0 (c) fast retransmit (d) RST MCQ
9. DHCP's four messages: (a) SYN/ACK/FIN/RST (b) DORA (c) Q/R/A (d) GET/PUT/POST/DEL MCQ
10. IP pings but names don't resolve → broken: (a) router (b) DNS (c) NAT (d) switch MCQ

Section B — Short Answer (3 × 6 = 18 marks)

11. Encapsulate "the packet leaves my laptop" — name the data unit and addressing at L4, L3, L2. 3 M
12. Differentiate circuit and packet switching (four points) and say why the Internet chose packets. 3 M
13. Why do MAC addresses change at every hop while IP addresses don't (until NAT)? 3 M
14. State split horizon and poison reverse; why do they cure count-to-infinity? 3 M
15. Idempotent vs safe methods — classify GET, POST, PUT, DELETE. 3 M
16. Explain the NAT state table's role for three devices sharing one public IP. 3 M

Section C — Long Answer (7 × 6 = 42 marks)

17. Compute: 1500-B packet over 3 store-and-forward links — 10 Mbps/600 km, 1 Mbps/30 km, 10 Mbps/1200 km, $s = 2 \times 10^8$ m/s. Per-link d_{trans} , d_{prop} , total one-way delay. 7 M
18. VLSM-allocate 192.168.20.0/24: Sales 100, Tech 50, Admin 20, Guest 10, WAN 2. Ranges + broadcasts, largest first. 7 M

19. Trace Dijkstra from Delhi on the 5-city graph (Lecture 5): expansion order, final costs, path to Chennai.

7 M

20. Count-to-infinity: diagram, the numbers after B-C fails, all four cures.

7 M

21. DNS resolution for `www.google.com` — every cache, server and referral, recursive vs iterative.

7 M

22. Three-way handshake + four-way close with seq/ack numbers; why `TIME_WAIT`?

7 M

23. The complete HTTPS journey (9 phases) with one protocol + one OSI layer per phase.

7 M

MOCK EXAM — ANSWER KEY

MCQ: 1-c · 2-c · 3-b · 4-b · 5-b · 6-c · 7-c · 8-c · 9-b · 10-b

11. Segment (ports) → Packet (IPs) → Frame (MACs); headers added going down.

12. Reserved path vs store-and-forward · FDM/TDM vs statistical sharing · guaranteed vs queueing · wasteful vs efficient; Internet = bursty traffic → packets.

13. L2 frames are local-only — rebuilt per hop; IP is the end-to-end identity routers forward on; NAT rewrites at the edge only.

14. Never re-advertise a route out the interface it came from (silence); poison reverse advertises it as ∞ explicitly — the false "path through you" never exists, so the count never starts.

15. GET safe+idempotent · POST neither · PUT idempotent not safe · DELETE idempotent not safe.

16. Each flow owns a translated port: (priv IP:port ↔ pub IP:port); replies look up by port; no entry → drop (accidental firewall).

17. $d_{trans} = 1.2 / 12 / 1.2 \text{ ms}$; $d_{prop} = 3 / 0.15 / 6 \text{ ms}$;

total = 23.55 ms ($L=12000$ bits).

18. Sales .0/25 (bc .127) · Tech .128/26 (bc .191) · Admin .192/27 (bc .223) · Guest .224/28 (bc .239) · WAN .240/30 (bc .243).

19. Order Delhi(0) → Hyderabad(2) → Bangalore(4) → Mumbai(5) → Chennai(8); Delhi→Chennai = Delhi→Hyd→Bang→Chen (8).

20. B sets $C=\infty$; A's stale "C=2" arrives → B:3 → A:4 → ... to 16; cures: split horizon, poison reverse, triggered updates, holddown.

21. browser → OS → hosts → resolver cache → root (referral .com) → TLD (referral ns1) → authoritative (A record, TTL) → cached & returned. Recursive for the client; iterative by the resolver.

22. $SYN(x) \rightarrow SYN-ACK(y, x+1) \rightarrow ACK(y+1)$; close $FIN(m) \rightarrow ACK(m+1) \rightarrow FIN(n) \rightarrow ACK(n+1)$; `TIME_WAIT` $2 \times MSL$ for a lost final ACK + duplicate expiry.

23. cache(L7 app) → DNS(L7/UDP) → ARP(L2) → TCP(L4) → TLS(L6) → HTTP(L7) → LB/app → NAT+routing(L3) → close(L4). Any faithful ordering with layers = full marks.

7 Glossary — Say It Like You Mean It

Term	One-line definition	Term	One-line definition
Protocol	agreed format + sequence + actions	Socket	(IP, port) endpoint of a connection
Encapsulation	wrapping data in per-layer headers going down	Three-way handshake	SYN → SYN-ACK → ACK
MTU	max frame payload (Ethernet 1500 B)	Sliding window	receiver-advertised unacked byte budget
Fragmentation	splitting oversized IP packets; reassembled at destination	Fast retransmit	resend on 3 duplicate ACKs
CIDR	/N prefix notation replacing classes	TTL	hop budget; 0 → drop + ICMP
VLSM	different prefix lengths per subnet	LPM	longest matching prefix wins
NAT/PAT	private IP:port ↔ public IP:port table	Convergence	all routers agree on tables
DORA	DHCP Discover Offer Request Ack	Count-to-infinity	DV looping after link failure
Gratuitous ARP	self-ARP: conflict check / cache refresh	LSA / LSDB	link-state advert / its database
APIPA	169.254.x.x = DHCP failed	AS / ASN	one admin's network / its number
Loopback	127.0.0.1 — never leaves the host	BGP hijack	more-specific prefix theft (RPKI fights it)
Throughput	achieved end-to-end rate = min link	SNI	hostname in ClientHello → right certificate
RTT / mdev	round-trip time / its jitter	ETag / 304	fingerprint / not-modified revalidation
Store-and-forward	router receives fully, then relays (+L/R)	TTFB	request → first response byte
PMTUD	DF-bit + ICMP discovery of path MTU	TLS 1.3	1-RTT encrypted channel, DH forward secrecy

THE NIGHT-BEFORE CARD — IF YOU READ NOTHING ELSE

Layers: PDNTSPA · App/Trans/Internet/NetAccess · Segment-Packet-Frame.

Delay: proc+queue+trans(L/R)+prop(d/s) · throughput=min · $\rho < 1$.

Routing: LPM longest wins · TTL-1 · RIP 15 hops · OSPF floods+Dijkstra · BGP AS_PATH policy.

TCP/UDP: SYN SYN-ACK ACK · FIN×2 · 3 dup-ACK retransmit · window=0 pause · UDP 8 B for DNS/video/games.

Troubleshoot: ladder L1→L7 · IP pings but name doesn't = DNS · 127.0.0.1 bind = invisible · CLOSE_WAIT = bug.

Subnet: IP AND mask = network · $2^h - 2$ hosts · /24→.0 /25→.128 /26→.192 /27→.224 /28→.240 /29→.248 /30→.252.

Graphs: BFS queue hops RIP · DFS stack cycles · Dijkstra heap cost OSPF · Bellman-Ford negatives · MST ↔ STP.

DNS/HTTP: caches→root→TLD→auth · GET safe · 401 vs 403 · cookies = state over stateless.

Local net: ARP bcast Q/unicast A · DORA · NAT port table · home router = 5 devices.

Journey: cache→DNS→ARP→TCP→TLS→HTTP→server→return→close · 2nd visit ≈ cache+keep-alive.